

Task 3:- Investigating the System

Now your supervisor wants you to gather basic information about the system you're working on.

Your Next Assignment:-

When you opened `mission-brief.txt`, it contained new message:

Great job, finding your way around the terminal. Your next assignment is to collect a small system report.

& who you're logged in as & The kernel version & Total disk space & The name of Linux distribution.

Step 1:- "Who are you logged in as?"

Whoami

ubuntu@tryhackme: ~\$ whoami } This prints your current username.

ubuntu

Step 2:- "What system are you on?"

To see details about operating system, kernel version and architecture, use: `uname -a`

```
ubuntu@tryhackme:~$ uname -a
Linux tryhackme 4.15.0-101-generic #101-Ubuntu SMP Mon Sep 2 13:48:07 UTC 2024 x86_64 x86_64 GNU/Linux
```

This gives a full line of system information - great for understanding exactly what environment you're working in.

Breakdown of information:-

linux:- The system is running the linux kernel. | tryhackme:- The hostname (the computer's name)

<REDACTED>-aws:- The kernel version installed on machine | x86_64:- The hardware platform (also 64 bit).

GNU/Linux:- The operating system type (linux kernel + GNU tools).

If you only wanted operating system name, you can try:- `uname`

ubuntu@tryhackme:~\$ uname } But for now `uname -a` gives full picture.

linux

Step 3:- Check Disk and Storage Info

In real world environments, you'll often need to check disk usage or available space, especially

before running tools or scripts in case. A simple command with readable output is `df -h`

```

df -h Command
ubuntu@tryhackme:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        12G  2.1G  9.9G   17% /
tmpfs            1.9G   0  1.9G   0% /dev/shm
tmpfs            774M  1.2M  773M   1% /run
tmpfs            5.0M   0  5.0M   0% /run/lock
tmpfs            387M  192K  387M   1% /run/user/1000
tmpfs            387M  172K  387M   1% /run/user/114

```

The -h means "human readable", it shows size like 2G or 500M instead of long bytes only numbers.

Breakdown of information! —

- * /dev/root :- is the main disk of system with --G total, 2G used, (RED ACTED) G free, and is 17% full.
- * tmpfs :- entries are temporary file system stored in RAM, not physical disk.
- * /dev/shm :- is shared memory area with 1.9G available and 0 used.
- * /run/user/114 :- is similar temporary storage for another system, also 387M total and mostly empty.

Step 4 :- Read a system file

Linux stores configuration and informational files in /etc directory.

To practice navigation and reading files, head into /etc by running `cd /etc` and then list what's inside: `ls`

```

cd Command
ubuntu@tryhackme:~$ cd /etc
ubuntu@tryhackme:/etc$ ls
ImageMagick-6  cloud          firefox        hp             logcheck       opt
ModemManager  compizconfig  fonts          ifplugd        login.defs     os-release
NetworkManager console-setup  fstab          init           logrotate.conf overlayroot.conf
...
-----
chatscripts   ethertypes    hosts.deny     localtime      openvpn        resolv.conf
ubuntu@tryhackme:/etc$

```

Let's now use `cat` to read the `os-release` file that almost every Linux system contains: `cat os-release`, as shown below:

```

cat Command
ubuntu@tryhackme:/etc$ cat os-release
PRETTY_NAME="Ubuntu 24.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.1 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo

```

This file provides details about Linux distribution that are often clearer than those provided by `uname` command.

Wrapping up your first day :- you navigated to file system, searched for hidden files, and gathered system information, solid progress for your first step into Linux. Before you finish your day, Supervisor has left one final task to test everything you've learned so far.